

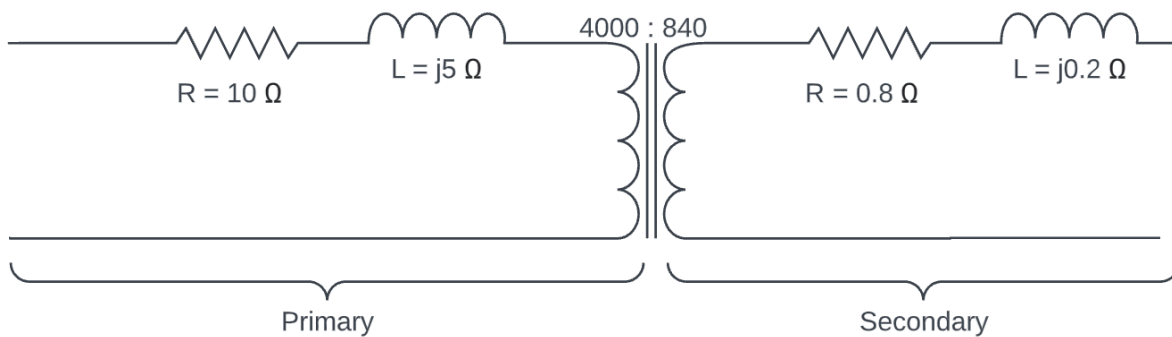
ECE 3113 - Energy Conversion

Homework 4

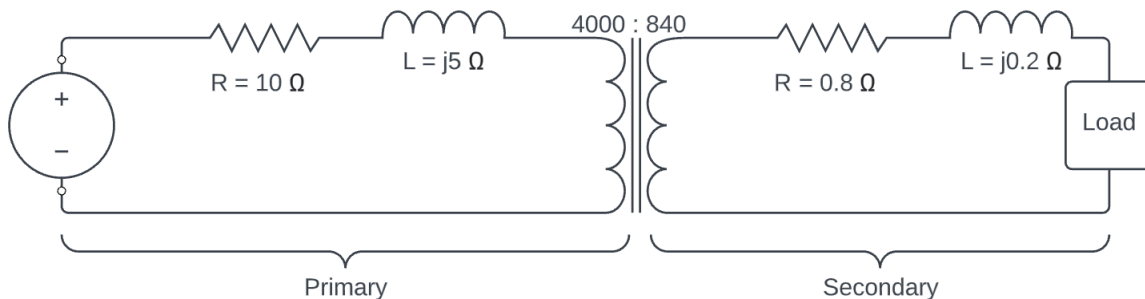
(Due on Monday 11/06/2023)

1. A 20 kVA single phase transformer 4000/840V has primary resistance and reactance of $10\ \Omega$ and $5\ \Omega$ respectively, while the resistance and reactance of secondary winding is $0.8\ \Omega$ and $0.2\ \Omega$ respectively. Using the simplified Equivalent circuit, Calculate:

a) The parameter referred to high voltage side and draw the equivalent circuit



b) The approximate value of secondary voltage at full load of 0.7 lagging power factor, suppose the voltage supply is $1500 \angle 0^\circ\text{V}$



2. A simple power system shown in figure below Contains a 250 V generator connected to an ideal 1:10 step up transformer, a transmission line, an ideal 20:1 step-down transformer, and a load. Calculate the currents and voltage at primary and secondary of transformers.

